

# An Omitted Mode Is a Rare Rule: The Sampling-Verification Danger Law in Continuous Code World Models

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## Abstract

In the Code World Model (CWM) paradigm, an LLM synthesizes an executable world model that a classical planner searches, accepted when it matches sampled transitions. In discrete games, a companion paper showed this gate certifies the wrong thing: a model can pass at 100% transition accuracy and still lose systematically, following  $\text{danger} = \text{play\_cost} \times (1 - \text{rarity})^N$  (Aguilar Martín, 2026). This paper moves to continuous state spaces, where model-based RL treats world-model error as *pervasive*, not localized. The law transfers — its ingredients are measure-theoretic — and we prove smooth (Lipschitz) truth/model pairs cannot realize the localized verified-but-wrong geometry (Section 3): the premise is realizable exactly where systems are hybrid, so an omitted mode is a rare rule. On two 1D hybrid instruments the gate misses the omitted mode at the predicted  $(1 - r)^N$  rate and the mode-blind planner is exploited below random at every knob; a distrust-region planner collapses this to a bounded first-contact transient. With real LLM synthesis, every sample missing the mode yielded a certified, blind, exploited model; when the sample contained the mode, GPT-5.x repaired the exact rule (82/82) — unlike the discrete setting. But repair is geometry-dependent: on a 4D bi-modal instrument with 2D modes it inverts. Neither GPT-5.x size recovers the region rule in any of **156** mode-containing seeds (disc, zero-curvature square, 3×-budget guidance), and a Claude cross-family arm reproduces the same failure classes. The mechanism is a *template prior* over low-complexity regions (half-planes on the disc, discs on the square), not boundary curvature, so the law’s exhaustiveness is geometry-scoped; certification stays sound. Identifiability is a sample property; repair is model- and geometry-dependent. Localization — both the danger and the repair — is a representational property of *code*.

## 1 Introduction

The companion paper (Aguilar Martín, 2026) established, in discrete games, that transition accuracy on randomly sampled play-throughs is the wrong adequacy criterion for a world model used in planning, and quantified the failure:  $\text{danger} = \text{play\_cost} \times (1 - \text{rarity})^N$ , with the gate-miss factor exact under i.i.d. sampling. Its related-work discussion closed with a promissory note: the rare-rule gap is a localized, discrete failure that state-accuracy metrics mask by dilution — a point worth revisiting in continuous settings, where the model-based RL literature (Lambert et al., 2020; Hafner et al., 2020; Janner et al., 2019) treats world-model error as pervasive and compounding rather than localized and pivotal.

This paper is that revisit. The question is not whether continuous world models can be wrong — that literature is mature — but whether the specific *verified-but-wrong* geometry of the discrete result exists in continuous state spaces: a model that passes a sampling gate cleanly, is (in a precise sense) almost-everywhere correct, and is still exploited catastrophically by the planner that trusts it.

The answer is yes, and the continuous version of the story is in one respect cleaner than the discrete one. The discrete headline had two components: (a) a provable one (when the rare rule is absent from the sample, no learner can infer it — an identifiability event with probability exactly  $(1 - r)^N$ ) and (b) an empirical one (the LLMs tested did not infer the rule *even when it was present* — “translation, not inference”). In the continuous *one-dimensional* instruments the second component **vanishes** for the models where we measured it: GPT-5.x infers an omitted hybrid mode from a handful of boundary-crossing transitions, writing the exact global rule. This turns out to be *geometry-dependent*: on a third instrument, a 4D bi-modal patch field whose modes are 2D regions rather than 1D clamps, the residual returns — neither GPT-5.x size recovers the region rule from data (0/156 mode-containing seeds pooled over the disc, a zero-curvature square ablation, and a strongest-effort prompting/budget treatment), and a Claude cross-family arm on the same cell reproduces the same failure classes (Section 7.1), so “translation, not inference” comes back on two-dimensional boundaries even for the model that erased it on the clamps. What remains provable is exactly the gate-miss and identifiability propositions (Section 3): with the mode absent from the sample — the  $(1 - r)^N$  event — no learner can recover it from the sample alone (a prior or the specification could still supply it; Proposition 2). Sample-covered exactness is the acceptance criterion itself; that every accepted artifact’s *off-sample* content was also the true mode — with a single exception, a prior-injected phantom mode erring only where its sample is silent — is an empirical, code-inspected regularity for the models tested (GPT-5.x repairs every revealed clamp; Qwen stalls and the gate refuses its patches; Claude repairs most through a symmetry prior), not a theorem. The danger law here is not just exact; empirically, for the models tested, it is exhaustive *on the 1D instruments* — and Section 7.1 shows that this exhaustiveness is itself geometry-scoped, failing on a 2D circular mode.

Contributions:

1. **Theory transfer, and one new piece.** The gate-miss proposition, the identifiability proposition, and the play-cost upper bound via query-hit mass transfer to continuous state spaces with only notational change — they are measure-theoretic (Section 3) — and a corollary ties the bound to the paper’s play\_cost normalization (Corollary 1). The genuinely discrete ingredient of the companion paper is the *localization premise*, and we prove it requires unbounded local Lipschitz structure: for  $L$ -Lipschitz truth and model differing by  $\eta$  at a state-action point, the disagreement region contains a metric ball of radius  $(\eta - \varepsilon)/2L$  in  $S \times A$  (Proposition 4). Hybrid mode boundaries are exactly where this obstruction disappears — an omitted mode is a rare rule.
2. **Two minimal hybrid instruments and two planner families** — cart-with-wall (linear off-mode plant) and pendulum-with-stop (nonlinear) — on which the full discrete phenomenology is measured with the same harness and no per-instrument re-calibration: the threshold law with the whole  $(1 - r)^N$  elbow inside the knob sweep, knob-invariant play\_cost  $\approx 1$  under random-shooting MPC, and the reach mechanism in its cleanest form (Section 4). A fixed CEM planner occupies the bound’s other branch: lower imagined mode-crossing and near-zero play cost on all 11 rows. Three instrument-design lessons of independent value are recorded (Section 2.3).
3. **A pinned-integrator contract and a measured axis separation.** The pinned-integrator synthesis contract makes correct code float-exact — a property of code exactness independent of the gate’s tolerance — so the synthesis gate (Section 7) can run at  $\varepsilon = 10^{-9}$ . Separately, at a deployment-realistic  $\varepsilon = 0.01$  (Section 5), the gate rejects supra-tolerance global bias on every rollout, passes sub-tolerance bias harmlessly, and misses the hard mode at a rate

that closely tracks the predicted  $(1 - r)^N$  (within sampling noise at gate scale; Section 5). A smooth localized perturbation of comparable rarity has  $\approx 0$ , at higher amplitude *negative*, play cost.

4. **A measured planner-side mitigation** (Section 6): distrust-region replanning — the planner checks its model’s predictions against observed transitions and fences the refuted predictions — collapses the pinned-forever, below-random exploitation to a bounded first-contact transient on all 11 knob rows across both hand-written instruments, at zero cost when the model is right (bit-identical to plain MPC, tested). The fence exploits the hard-boundary structure of these modes (Section 6); the gate-side law is untouched: the gate still certifies a wrong model.
5. **The synthesis result: danger collapses to pure identifiability** (Section 7). Real-LLM arms (GPT-5.x, 20 seeds/cell) with the identifiability event logged per seed: mode absent  $\Rightarrow$  20/20 verified-blind-exploited (regret 0.999; Wilson lower bound 0.72 per size — the 20 are 10 shared samples  $\times$  2 sizes, Section 7); mode present  $\Rightarrow$  repaired to the exact global rule (both sizes 10/10, large in  $\leq 1$  iteration), never a wrong artifact accepted in the GPT-5.x arms. Cross-family spot-checks in two more families reproduce the blind-exploited event in every family but diverge on repair — Qwen repairs none (superstitious patches the gate refuses), Claude (agent-relayed) repairs most through a symmetry prior that can certify an invented mode unfalsifiable by its gate sample — so repair-from-data is model-dependent in mechanism, while identifiability is a property of the sample, not the model. The whole result replicates on a second, nonlinear instrument (a pendulum with an angular hard stop): GPT-5.x repairs 62/62 mode-present seeds and every mode-absent seed is blind and exploited. On a third, 4D bi-modal instrument (PatchField2D) the repair finding *inverts*: neither GPT-5.x size recovers the circular mode from data (0/76 mode-containing seeds), reducing the disc to a half-plane, and a Claude cross-family arm fails the same way. Two ablations then pin the mechanism: a *zero-curvature square* is not repaired either (0/40 — with the inverse error, discs written on square evidence), and region-first guidance at  $3\times$  budget eliminates the half-plane yet yields hull fits of the one-sided contact evidence (0/40) — **0/156 pooled**. Repair-from-data is geometry-dependent via a *template prior* over region forms, not curvature, and the law’s empirical exhaustiveness is correspondingly scoped — while the all-or-nothing gate stays sound, certifying no partial artifact (Section 7.1).
6. **Smooth learners cannot localize** (Section 8): a closed-form linear fit on mode-free data passes both gates fully blind (identifiability is learner-independent); on mode-containing data it is tilted twelve orders of magnitude off-mode by four contact rows in 3200 while still missing the mode. Both the danger geometry and the repair capability are representational properties of code.

Scope up front: three instruments (two 1D — cart, pendulum — plus a 4D bi-modal PatchField2D), two base planner families (random-shooting MPC and CEM, one fixed configuration each), 20 seeds per headline synthesis cell across the instruments (both GPT-5.x sizes; plus the caught cell on the pendulum and two bi-knob cells on PatchField2D), a 3-seed cross-family spot-check per 1D instrument (not full sweeps), and a probe-grade MLP. Section 10 gives the honest assessment and the full seed accounting.

## 2 The instruments

This section specifies the hand-written hybrid instruments and the planner that every later experiment reuses, and records the design lessons learned while calibrating them.

### 2.1 Cart-with-wall

State  $(x, v)$ ; action  $a \in [-a_{\max}, a_{\max}]$ ; semi-implicit Euler with fixed  $dt$ :

$$a \leftarrow \text{clamp}(a), \quad v' = v + (\text{gain} \cdot a - \text{drag} \cdot v) dt, \quad x' = x + v' dt,$$

with defaults  $dt = 0.1$ ,  $\text{gain} = 3$ ,  $\text{drag} = 0.3$ ,  $a_{\max} = 1$ . The **hybrid mode** is an inelastic wall at  $x_{\text{wall}}$ : if  $x' \geq x_{\text{wall}}$ , the next state is exactly  $(x_{\text{wall}}, 0)$ . The **blind model** is the same code path with the wall branch removed — the hand-written on-manifold proxy for a CWM synthesized from a spec that omits the wall, *bit-exact off-mode by construction* (tested: exact float equality on wall-free trajectories). We call such a model **mode-blind** (the canonical term; instrument-specific synonyms like “wall-blind”, or “blind” for short, mean the same thing). Reward is two sigmoid plateaus: a small reachable one on the left (0.3, at  $x \leq -6$ ) and a large one on the right (1.0, at  $x \geq 12$ ) whose approach every swept wall position blocks. Episodes are 80 steps from  $x_0 \sim U(-0.5, 0.5)$ ,  $v_0 = 0$ .

The design requirements mirror the companion paper’s material-at-cap instrument (paper 1’s discrete instrument: a rare resource-cap rule likewise omittable from sampled play): (a) random rollouts rarely fire the mode — the wall position is the **rarity knob** ( $r$  sweeps  $0.331 \rightarrow 0.0020$  over  $x_{\text{wall}} \in [2, 10]$ ; Table 1); (b) the omission is *exploited*, not merely mispredicted — the wall-blind model predicts coasting through the wall toward the large plateau, so the planner it advises drives right and is pinned; (c) the truth planner’s optimal play differs qualitatively — it goes left.

The second instrument (pendulum with a hard angular stop; nonlinear gravity term) is described with its results in Section 4.1, and the third (PatchField2D, a 4D bi-modal patch field) in Section 4.2.

### 2.2 Planner and play\_cost

The planner is model-predictive control (MPC) by random shooting (Nagabandi et al., 2018; Chua et al., 2018): at each step, sample candidate action sequences (piecewise-constant blocks plus the three constant sequences  $\{-a_{\max}, 0, +a_{\max}\}$ ), roll each out on the *model*, take the best first action, replan. The planner is a deterministic function of its model’s responses and the seed, so the play-cost upper bound via query-hit mass applies verbatim (Section 3; Corollary 1 ties that bound to the normalization below). Single-agent control makes `play_cost` a normalized regret, cleaner than a two-player arena (no opponent confound):

$$\text{play\_cost} = \frac{J_{\text{truth}} - J_{\text{model}}}{J_{\text{truth}} - J_{\text{rand}}},$$

where  $J_{\text{truth}}$ ,  $J_{\text{model}}$ , and  $J_{\text{rand}}$  are the returns of the truth-planner, the model-planner (written  $J_{\text{blind}}$  in the tables when the model is the blind one), and the uniform-random policy, all measured in the true environment on paired seeds. The blind planner can score *below random* (it is actively exploited), so the normalized value can exceed 1; we report it unclamped.

### 2.3 Three instrument-design lessons

Recorded because they will bite any continuous CWM study:

1. **I.i.d. per-step candidate sampling silently removes the mode from imagination.** With i.i.d. candidates, imagined displacement is diffusive; no sampled sequence reaches distant reward within the horizon, so the truth model and the blind model rank all candidates *identically* and the wall never enters imagination — the two arms become indistinguishable not because the models agree but because the planner never queries where they differ. Piecewise-constant blocks plus constant candidates fix it. The planner’s *query* distribution, not just its trajectory distribution, is part of the instrument.
2. **Point (Gaussian) reward lodes demand braking finesse that random shooting lacks;** sigmoid plateaus remove the parking problem and give clean imagined-value margins in both directions.
3. **The plant’s drag time-constant must sit well inside the planning horizon,** or no arm can act on the reward at all (our first calibration had  $\tau = 1/\text{drag} = 10\text{ s}$  against a 3 s horizon; nothing moved).

### 3 Theory: what transfers, what changes, what is new

Notation: a verification gate draws  $N$  i.i.d. rollouts from the gate policy  $\rho$  (uniform-random actions from the initial-state distribution) and accepts a model  $\hat{f}$  if it matches the truth  $f$  within  $\varepsilon$  in sup-norm on every visited transition.

**Measure-space setup.** Fix a horizon  $T$ . A trajectory is a point of  $(S \times A)^T$ , equipped with the product Borel  $\sigma$ -algebra. The gate policy  $\rho$  together with the transition kernel induced by  $f$  defines a trajectory law  $P_\rho$  on this space (the initial-state distribution pushed forward through  $\rho$  and  $f$ ). The query measure  $\mu_{\text{query}}(E)$  (the *query-hit mass*) is the probability, under the planner’s trajectory law, that at least one model query during the episode lands in  $E$ . We assume throughout that the critical region  $R$ , the disagreement region  $E$ , and the reward and dynamics maps are Borel measurable, so every probability below is well defined. With this in place the discrete arguments of [Aguilar Martín \(2026\)](#) transfer with only notational change.

**Proposition 1** (gate miss; transfers verbatim). *Let  $R$  be any measurable set of rollouts (“the critical event”; here: the rollout fires the wall mode) with  $r = P_\rho(R)$ . The probability that  $N$  i.i.d. gate rollouts all avoid  $R$  is exactly  $(1 - r)^N$ .*

Nothing in the companion paper’s proof uses discreteness; the event is  $\text{Bernoulli}(r)$  and the draws are i.i.d.

**Proposition 2** (identifiability; transfers verbatim). *Condition on the miss event of Proposition 1. Let  $M \subseteq S \times A$  be the mode region (for the cart the clamp fires depending on  $(s, a)$  through  $x' \geq x_{\text{wall}}$ ), and let  $\hat{f}_1, \hat{f}_2$  be any two models that agree on  $(S \times A) \setminus M$ . On the miss event the sample visits no transition in  $M$ , so  $\hat{f}_1$  and  $\hat{f}_2$  produce identical outputs on every sampled input; hence every sample-measurable score (the gate, a likelihood, a refinement objective) is constant across the two, and no such score can distinguish them — preference for the correct one must come from the prior or the specification. This holds for any learner — LLM, linear regression, MLP — a point Section 8 instantiates empirically.*

**Proposition 3** (play-cost upper bound; transfers verbatim). *Assume returns are normalized to  $[0, 1]$  (WLOG, by rescaling  $J \mapsto (J - J_{\min})/(J_{\max} - J_{\min})$ ; equivalently carry an explicit factor  $J_{\max} - J_{\min}$  throughout). For any planner that is a deterministic function of model responses and a*

seed,  $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)$ , where  $E$  is the disagreement region and  $\mu_{\text{query}}$  the probability (the query-hit mass) that the planner queries its model on  $E$  during an episode.

*Proof sketch (coupling).* Couple the two runs on a common seed. On the event that no query lands in  $E$ ,  $f$  and  $\hat{f}$  return identical responses to every query the planner issues (they agree off  $E$ ), so with shared rng the two runs select the same action sequence and traverse the same states, realizing the same return. The realized returns can therefore differ only on the complementary event, of probability  $\mu_{\text{query}}(E)$ , where normalized returns differ by at most 1; taking expectations gives  $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)$ . MPC’s imagined rollouts are the queries. (The argument is the coupling of [Aguilar Martín \(2026\)](#), recorded here rather than deferred.)

The raw  $J$  reported in the tables (e.g.  $J_{\text{truth}} = 17.77$ ) is this normalized quantity rescaled by  $J_{\text{max}} - J_{\text{min}}$ , so the bound is a statement about the normalized return. The paper’s reported `play_cost`, however, divides by  $J_{\text{truth}} - J_{\text{rand}}$  (Section 2.2), *not* by  $J_{\text{max}} - J_{\text{min}}$ ; the following corollary states the relationship rather than silently identifying the two.

**Corollary 1** (play\_cost saturation). *With  $|\text{play\_cost}| = |J(f) - J(\hat{f})| / (J_{\text{truth}} - J_{\text{rand}})$ , play\_cost as defined (signed) in Section 2.2, Proposition 3 in raw units ( $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E) (J_{\text{max}} - J_{\text{min}})$ ) gives*

$$|\text{play\_cost}| \leq \mu_{\text{query}}(E) \frac{J_{\text{max}} - J_{\text{min}}}{J_{\text{truth}} - J_{\text{rand}}},$$

*so the bound reads  $|\text{play\_cost}| \lesssim \mu_{\text{query}}(E)$  exactly when  $J_{\text{rand}} \approx J_{\text{min}}$  and  $J_{\text{truth}} \approx J_{\text{max}}$  — the random policy near the reward floor, the truth planner near the ceiling. Both instruments are in this regime: on the cart  $J_{\text{rand}} = 0.53$  against a pinned  $J_{\text{blind}} \approx 0$  and  $J_{\text{truth}} = 17.77$ ; on the pendulum  $J_{\text{rand}} = 0.06$  and  $J_{\text{truth}} = 20.08$ . The blind planner queries the wall region in every episode, so  $\mu_{\text{query}}(E) \approx 1$  and the normalized bound is saturated —  $\text{play\_cost} \approx 1$ , as observed in Sections 4–5.*

**What does not transfer.** The companion paper’s coverage certificates enumerate finite information-set spaces. Continuous state spaces admit no such enumeration; covering-number analogues under Lipschitz assumptions are left open (Section 10).

**What is new: the localization premise is a theorem-shaped obstruction.**

**Proposition 4** (smoothness forbids localized error). *Let  $f, \hat{f} : S \times A \subseteq \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}^d$  be  $L$ -Lipschitz on the joint state-action space in the sup-metric on  $S \times A$ , with  $L \in [0, \infty)$  (i.e.  $L = \max(\text{Lip } f, \text{Lip } \hat{f})$ ). Suppose  $\|f(s_0, a_0) - \hat{f}(s_0, a_0)\|_\infty = \eta$  at some  $(s_0, a_0) \in S \times A$ . Then for any tolerance  $\varepsilon < \eta$ , the disagreement region  $E_\varepsilon = \{(s, a) \in S \times A : \|f(s, a) - \hat{f}(s, a)\|_\infty > \varepsilon\}$  contains the open metric ball  $B((s_0, a_0), \frac{\eta - \varepsilon}{2L}) \cap (S \times A)$  (the intersection matters only when  $(s_0, a_0)$  lies near  $\partial(S \times A)$ ). If instead  $L = 0$  then  $f - \hat{f}$  is constant and  $E_\varepsilon$  is all of  $S \times A$ .*

*Proof.*  $g = f - \hat{f}$  is  $2L$ -Lipschitz on  $S \times A$ , so for  $\|(s, a) - (s_0, a_0)\|_\infty < (\eta - \varepsilon)/2L$  we have  $\|g(s, a)\|_\infty \geq \eta - 2L \|(s, a) - (s_0, a_0)\|_\infty > \varepsilon$ ; the inequality is strict, so the ball is open. The  $L = 0$  case is immediate:  $g \equiv g(s_0, a_0)$  with  $\|g(s_0, a_0)\|_\infty = \eta > \varepsilon$ .  $\square$

The disagreement region  $E_\varepsilon$  lives in the same joint state-action space  $S \times A$  as the query region  $E$  of Proposition 3, so the ball is *directly* a lower bound on the query-relevant region the planner can hit — no fixed-action slice argument is needed to bring the two together. (Restricting to a fixed action  $a_0$  recovers the state-slice form: the open ball of radius  $(\eta - \varepsilon)/2L$  in  $S \times \{a_0\}$  lies in  $E_\varepsilon$ , the version convenient when reasoning about a single instrument’s mode.)



Read as its contrapositive: a model error that is *large somewhere* (large enough to matter at play) and *invisible at tolerance  $\varepsilon$  outside a metrically tiny region* requires  $L$  large — in the limit, a discontinuity. This is exactly the sense in which the discrete localization premise (“the wrong model is exact off the rule region”) is discrete: it is unsatisfiable by smooth truth/model pairs, up to the ball-radius quantification. Two consequences structure the paper: (i) the natural continuous home of the danger geometry is **hybrid dynamics** — contacts, hard stops, saturations, regime switches — where the truth itself has unbounded local Lipschitz constant across a mode boundary (our wall:  $v$  jumps to 0), and an omitted mode is precisely a rare rule; (ii) the premise is *representationally* available to programs (an omitted or added branch is bit-exact off the omitted case) and not to smooth learned models, a distinction Section 8 measures.

*Remark 1* (metric, not measure). The ball in Proposition 4 is metric, and its *probability* under the gate’s visitation measure can still be small — smoothness bounds how spatially concentrated an error can be, not how often the gate visits it. The proposition removes the *exact* localization premise for smooth pairs; the  $(1 - r)^N$  mechanism itself is representation-independent and applies to any critical region of small visitation measure.

## 4 The mechanism and the threshold law

This section measures the danger law’s full phenomenology on the hand-written instruments, before any LLM enters the loop. All numbers CPU-only, with the hand-written blind model as the on-manifold proxy (`scripts/continuous_reach.py`; 3000 rarity rollouts and 20 MPC episodes/arm per knob; Wilson 95% CIs (Wilson, 1927) in the versioned results).

| $x_{\text{wall}}$ | rarity | $J_{\text{truth}}$ | $J_{\text{blind}}$ | $J_{\text{rand}}$ | play_cost | blind hit | truth hit | d@20  | d@40  | d@80  |
|-------------------|--------|--------------------|--------------------|-------------------|-----------|-----------|-----------|-------|-------|-------|
| 2                 | 0.3313 | 17.77              | 0.00               | 0.53              | 1.031     | 1.00      | 0.00      | 0.000 | 0.000 | 0.000 |
| 3                 | 0.2193 | 17.77              | 0.00               | 0.53              | 1.031     | 1.00      | 0.00      | 0.007 | 0.000 | 0.000 |
| 4                 | 0.1430 | 17.77              | 0.00               | 0.53              | 1.031     | 1.00      | 0.00      | 0.047 | 0.002 | 0.000 |
| 5                 | 0.0843 | 17.77              | 0.00               | 0.53              | 1.031     | 1.00      | 0.00      | 0.177 | 0.030 | 0.001 |
| 6                 | 0.0503 | 17.77              | 0.00               | 0.53              | 1.031     | 1.00      | 0.00      | 0.367 | 0.131 | 0.017 |
| 8                 | 0.0127 | 17.77              | 0.02               | 0.53              | 1.030     | 1.00      | 0.00      | 0.798 | 0.619 | 0.372 |
| 10                | 0.0020 | 17.77              | 0.94               | 0.53              | 0.977     | 1.00      | 0.00      | 0.938 | 0.901 | 0.832 |

Table 1: The danger curve on the wall-position knob. “blind/truth hit” = fraction of episodes in which that planner’s trajectory fires the wall mode;  $d@N = \text{play\_cost} \cdot (1 - r)^N$ .

Readings: **the threshold law again** — danger  $\approx 0$  while the wall sits inside the random-rollout envelope, rises through the elbow, and plateaus at full `play_cost`, with the entire elbow inside the sweep (Figure 1). **play\_cost is knob-invariant and  $\approx 1$**  (exceeding 1 at every knob except the largest, where far-plateau leakage gives 0.977): the blind planner is not merely uninformed but *exploited* — MPC on the wall-less model plans into the phantom region, is pinned at the wall (final  $x = x_{\text{wall}}$  exactly, contact rate 1.00 at every knob), and replans the same doomed plan every step, for the entire episode, scoring below random (0.00 vs 0.53). The 0.977 at  $x_{\text{wall}} = 10$  is the sigmoid tail of the far plateau leaking  $J_{\text{blind}} = 0.94$ ; the mechanism is unchanged. **The reach mechanism** appears in its cleanest form (Figure 2), with the query/trajectory distinction now visible. Gate-miss exactness is re-verified in-tests and again at gate scale in Section 5.

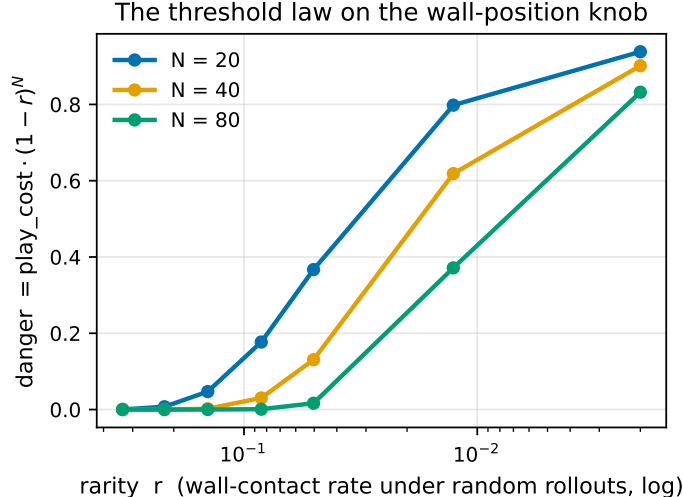


Figure 1: The threshold law on the wall-position knob (Table 1 rendered):  $\text{danger} \approx 0$  while the wall is inside the random envelope, rises through the elbow, plateaus at full  $\text{play\_cost}$ ;  $N$  shifts the threshold.

#### 4.1 Robustness: the same law on a nonlinear plant

The cart’s off-mode dynamics are linear, which is convenient for Section 8 but invites the worry that the phenomenology depends on it. A second instrument — a pendulum (gravity term  $\sin \theta$ ,  $\theta = 0$  hanging down) with a hard angular stop, same interface, same MPC, same two-plateau reward on  $\theta$  — reproduces the identical picture with *no* re-calibration (rarity is natural here: gravity confines the random walk near the bottom, so climbing to the stop is rare):

| $\theta_{\text{stop}}$ | rarity | $J_{\text{truth}}$ | $J_{\text{blind}}$ | play_cost | blind hit | d@40         |
|------------------------|--------|--------------------|--------------------|-----------|-----------|--------------|
| 0.8                    | 0.2970 | 20.08              | 0.01               | 1.002     | 1.00      | 0.000        |
| 1.0                    | 0.1277 | 20.08              | 0.03               | 1.002     | 1.00      | 0.004        |
| 1.2                    | 0.0527 | 20.08              | 0.05               | 1.000     | 1.00      | 0.115        |
| 1.4                    | 0.0193 | 20.08              | 0.12               | 0.997     | 1.00      | 0.457        |
| 1.6                    | 0.0073 | 20.08              | 0.26               | 0.990     | 1.00      | 0.737        |
| 2.0                    | 0/3000 | 20.08              | 1.23               | 0.942     | 1.00      | $\leq 0.942$ |

Table 2: Pendulum-with-stop: threshold law, knob-invariant exploitation (pinned at the stop in every episode at every knob), truth planner untouched by the mode. The mechanism does not care that the plant is nonlinear — only that the mode is hard and rare under the gate’s measure. Rarity is the fraction of 3000 random rollouts that fire the stop (Wilson 95% CIs for every row in the versioned JSON). At  $\theta_{\text{stop}} = 2.0$  *no* rollout fires it: that entry is the raw count rather than a point estimate of 0, and its d@40 is correspondingly an upper bound — at the Wilson upper rarity (0.0013) it would be 0.895.

#### 4.2 The danger law composes: a 4D bi-modal instrument

Both instruments so far are one hard boundary in a 2D state. **PatchField2D** closes both structural gaps at once — a 4D state and two independent modes — and asks what the 1D instruments cannot:



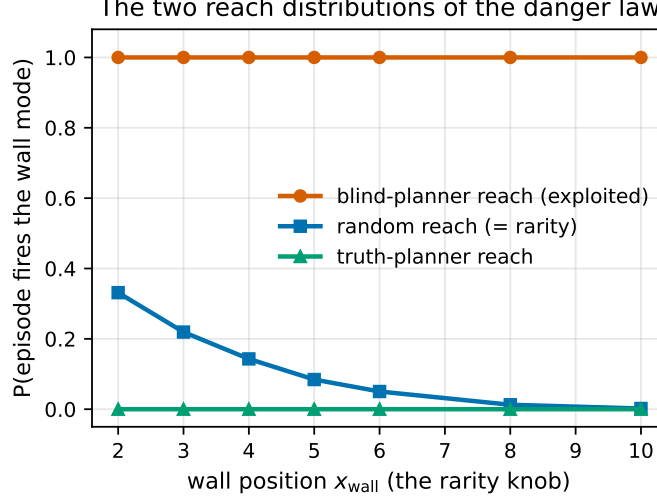


Figure 2: The two reach distributions of the danger law: the exploited planner’s mode reach is flat at 1.00 across the knob; random reach (= rarity) falls two orders of magnitude; the truth planner’s trajectory reach is 0.00 — the mode lives on its *query* distribution (Proposition 3), not its path.

does the danger law compose mode-wise? State  $(x, y, v_x, v_y)$ ; a *scalar* action  $a \in [-a_{\max}, a_{\max}]$  is mapped to a heading  $\phi = \pi \text{ clamp}(a)/a_{\max}$  so all planner machinery (which consumes scalar actions) is reused unchanged; semi-implicit Euler:

$$v'_x = v_x + (\text{gain} \cos \phi - \text{drag } v_x) dt, \quad v'_y = v_y + (\text{gain} \sin \phi - \text{drag } v_y) dt,$$

$x' = x + v'_x dt$ ,  $y' = y + v'_y dt$ . The two **modes** are circular sticky patches  $P_i = \text{disc}(c_i, R)$ : if  $(x', y') \in P_i$  the next state is  $(x, y, 0, 0)$  — the probe stops inelastically at its *previous* position with zero velocity (the 2D analogue of the wall clamp, and the structure the 2D mitigation exploits, Section 6.1). Reward is two radial sigmoid lodes, a small real one near the start and a large phantom one behind the patches; the mode-blind planner aims straight at the phantom and freezes at a patch edge. The two knobs are the patch centers’ distances along and off the start-to-lode corridor ( $k_1$  the near patch,  $k_2$  the far one); defaults are frozen once, never tuned per cell.

Table 3 sweeps the  $3 \times 3$  knob grid (`scripts/continuous_patch2d.py`; 600 rarity rollouts and 20 MPC episodes/arm per cell). The per-mode rarities separate cleanly ( $r_1 \in [0.085, 0.245]$ ,  $r_2 \in [0.0067, 0.0100]$ ),  $J_{\text{blind}} = 0$  at every cell (the blind planner freezes at a patch edge every episode;  $J_{\text{rand}} = 0.11$ ), and `play_cost` is knob-invariant at  $[1.005, 1.006]$  — the same below-random exploitation as the 1D instruments, now on a 4D plant. **The danger law composes per mode:** the per-mode gate-miss factors are  $(1 - r_i)^N$  and the joint factor is  $(1 - r_1)^N(1 - r_2)^N$ , tabulated as the d@40 columns. Two honest notes: the  $r_2$  knob is only weakly resolved at 600 rollouts (its trend is under-resolved, not flat), and the  $k_1=4$  one-hit bump in  $r_2$  is sampling noise rather than a knob effect.

### 4.3 A second planner family: play\_cost is planner-dependent

Proposition 3 predicts two branches: a blind model can change behavior only to the extent that the planner queries its disagreement region. Random-shooting MPC’s constant candidates reach the distant phantom plateau in imagination and produce `play_cost`  $\approx 1$ . We repeated the full 11-knob grid with a second base planner, the cross-entropy method (CEM; `scripts/continuous_cem.py`;

| $k_1$ | $k_2$ | $r_1$  | $r_2$  | $J_{\text{truth}}$ | play_cost | d@40 $P_1$ | d@40 $P_2$ | d@40 joint |
|-------|-------|--------|--------|--------------------|-----------|------------|------------|------------|
| 2     | 6     | 0.2450 | 0.0100 | 17.98              | 1.006     | 0.0000     | 0.6732     | 0.0000     |
| 2     | 7     | 0.2450 | 0.0083 | 18.02              | 1.006     | 0.0000     | 0.7200     | 0.0000     |
| 2     | 8     | 0.2450 | 0.0067 | 18.08              | 1.006     | 0.0000     | 0.7700     | 0.0000     |
| 3     | 6     | 0.1417 | 0.0083 | 18.57              | 1.006     | 0.0022     | 0.7197     | 0.0016     |
| 3     | 7     | 0.1417 | 0.0083 | 18.47              | 1.006     | 0.0022     | 0.7197     | 0.0016     |
| 3     | 8     | 0.1417 | 0.0067 | 17.72              | 1.006     | 0.0022     | 0.7699     | 0.0017     |
| 4     | 6     | 0.0883 | 0.0083 | 20.85              | 1.005     | 0.0249     | 0.7192     | 0.0178     |
| 4     | 7     | 0.0883 | 0.0100 | 19.52              | 1.006     | 0.0249     | 0.6727     | 0.0166     |
| 4     | 8     | 0.0850 | 0.0083 | 19.08              | 1.006     | 0.0288     | 0.7196     | 0.0206     |

Table 3: PatchField2D mechanism sweep ( $3 \times 3$  knobs; 600 rarity rollouts, 20 MPC episodes/arm;  $J_{\text{blind}} = 0$ ,  $J_{\text{rand}} = 0.11$  at every cell).  $\text{d@40 } P_i = \text{play\_cost} \cdot (1 - r_i)^{40}$  (per-mode danger);  $\text{d@40 joint} = \text{play\_cost} \cdot ((1 - r_1)(1 - r_2))^{40}$ . The law composes mode-wise. Full table (with  $J_{\text{blind}}$ ,  $J_{\text{rand}}$ , hit rates) in docs/EXPERIMENTS.md.

horizon 40, 5 iterations, 64 samples, elite fraction 0.125, minimum standard deviation 0.05; one fixed setting across both instruments). The crossing columns in Table 4 are the fraction of sampled imagined trajectories that cross the omitted boundary, measured for BOTH planners with one plan from the same paired initial state per episode seed (episode-accumulated CEM fractions, which tell the same story, are recorded in the results JSON).

| instrument | knob | pc MPC | pc CEM | contact CEM | crossing CEM | crossing MPC |
|------------|------|--------|--------|-------------|--------------|--------------|
| cart       | 2.0  | 1.031  | 0.000  | 0.00        | 0.0309       | 0.3865       |
| cart       | 4.0  | 1.031  | 0.000  | 0.00        | 0.0055       | 0.2453       |
| cart       | 6.0  | 1.031  | 0.000  | 0.00        | 0.0003       | 0.1483       |
| cart       | 8.0  | 1.030  | 0.000  | 0.00        | 0.0000       | 0.0773       |
| cart       | 10.0 | 0.977  | 0.000  | 0.00        | 0.0000       | 0.0369       |
| pendulum   | 0.8  | 1.002  | 0.009  | 0.70        | 0.1150       | 0.6392       |
| pendulum   | 1.0  | 1.002  | 0.025  | 0.25        | 0.0483       | 0.5530       |
| pendulum   | 1.2  | 1.000  | -0.011 | 0.00        | 0.0164       | 0.4672       |
| pendulum   | 1.4  | 0.997  | -0.021 | 0.00        | 0.0053       | 0.3842       |
| pendulum   | 1.6  | 0.990  | -0.021 | 0.00        | 0.0013       | 0.3039       |
| pendulum   | 2.0  | 0.942  | 0.000  | 0.00        | 0.0002       | 0.2158       |

Table 4: The same certified-blind models under two base planner families (20 paired episodes/row). “pc” is blind-model play\_cost within that planner family; crossing is the sampled imagined-boundary-crossing proxy for query-hit mass, one plan per episode seed from paired initial states for both planners. CEM stays near zero play cost and below MPC’s crossing fraction on every row.

CEM’s blind-model play cost lies in  $[-0.0213, 0.0248]$  on every row — and the seed-paired 95% t-interval includes zero on all 11 rows — while its imagined crossing fraction is strictly below MPC’s throughout. On the cart, truth- and blind-model CEM returns are identical and contact is zero everywhere. The nearest pendulum stops are an honest qualification: CEM contacts  $\theta_{\text{stop}} = 0.8$  in 70% of episodes and 1.0 in 25%, but does not enter MPC’s pinned, below-random regime; contact is zero from  $\theta_{\text{stop}} \geq 1.2$ . Contact is therefore not itself exploitation. This is the other measured branch of Proposition 3: the same certified-blind model is a landmine whose consequence depends on the planner’s query reach, operationalizing Section 2.3’s first lesson — if search does not discover the phantom, it cannot optimize toward it.

Two caveats prevent the wrong conclusion. CEM’s pendulum truth return varies from 15.36 to 16.46 (versus MPC’s 20.08), consistent with local optima; the comparison is blind-CEM against truth-CEM, not a claim that CEM is globally optimal. And limited reach is not knowledge or mitigation: a planner that misses a phantom distant reward can also miss a real one. This is one fixed CEM configuration, not a hyperparameter sweep.

**PatchField2D: the anticipated 2D competence gap did not materialize.** On the 4D bi-modal instrument (`scripts/continuous_cem_patch2d.py`) blind CEM is again *not* exploited: `play_cost` is  $-0.022/+0.017/+0.020$  at knobs (2, 6)/(3, 7)/(4, 8) with the seed-paired 95%  $t$ -interval including zero on all three rows, and CEM’s imagined crossing fraction is below MPC’s on every row ( $0.070 < 0.208$ ,  $0.027 < 0.149$ ,  $0.009 < 0.094$ ). Here CEM is *competent in aggregate* — the 2D competence gap we anticipated did not appear — but with per-seed variance (one truth-CEM episode at  $\approx 0.97$ ), and its blind contact rate is a uniform 0.05 (nonzero, unlike the cart’s 0.00, and without positive play cost). The same low-query-reach branch of Proposition 3, one more instrument.

## 5 Axis separation: the gate fails only where the law says it can

The classic continuous-model failure axis is pervasive sub-tolerance error; the danger law’s axis is a localized hard mode. A tolerance gate ( $\varepsilon = 0.01$ , deployment-realistic) must be shown to fail *only* on the second axis, and only at the  $(1-r)^N$  rate. Five arms, one table (`scripts/continuous_axes.py`; reveal-rarity =  $P(\text{a random rollout contains a transition where truth and model differ } > \varepsilon)$ , 2000 rollouts; pass@40 over 300 independent  $N=40$  gates; 20 MPC episodes/arm). For the hard mode, the mode-firing rarity of Tables 1–2 and the reveal-rarity here coincide as events — the mode error exceeds every deployment-realistic  $\varepsilon$  (the  $\varepsilon$ -sweep below), so a rollout fires the mode iff it reveals a disagreement — which licenses the shared symbol  $r$ .<sup>1</sup>

| arm                                            | reveal-rarity | $(1-r)^{40}$ | pass@40 measured | play_cost | d@40    |
|------------------------------------------------|---------------|--------------|------------------|-----------|---------|
| wall@4 omitted                                 | 0.1385        | 0.0026       | 0.003            | 1.031     | 0.0027  |
| wall@8 omitted                                 | 0.0125        | 0.6046       | 0.667            | 1.030     | 0.6227  |
| drag bias $\times 1.03$ (sub- $\varepsilon$ )  | 0.0000        | 1.0000       | 0.997            | 0.000     | 0.0000  |
| drag bias $\times 2.0$ (supra- $\varepsilon$ ) | 1.0000        | 0.0000       | 0.000            | 0.000     | 0.0000  |
| $C^\infty$ bump@4, amp 0.5                     | 0.1875        | 0.0002       | 0.000            | 0.000     | 0.0000  |
| $C^\infty$ bump@4, amp 1.0                     | 0.2085        | 0.0001       | 0.000            | -0.745    | -0.0001 |

Table 5: Axis separation at  $\varepsilon = 0.01$ ,  $N = 40$ ;  $d@N = \text{play\_cost} \cdot (1-r)^N$ , as in Table 1. Danger lives in one quadrant only: rare  $\wedge$  hard-mode.

Readings (Table 5, rendered as the danger quadrant in Figure 3): **gate exactness at gate scale** — measured pass@40 matches  $(1-r)^{40}$  in both wall rows (0.003 vs 0.0026; 0.667 vs 0.605, the prediction marginally below the 300-gate Wilson lower bound of 0.6115 — sampling noise at that gate count; see the  $\varepsilon$ -sweep note below); the proposition is the observed acceptance rate, not asymptotic decoration. **The gate polices the pervasive axis:** a global drag bias above tolerance is revealed on every rollout and never accepted; a sub-tolerance bias is accepted (0.997; the 0.3% is the extreme velocity tail) and is harmless at play. Verified-and-fine is a real cell, and the gate finds it. **Smoothness kills consequence, not detectability:** the  $C^\infty$  drag bump at the wall’s location has comparable rarity to the wall (0.19 vs 0.14 — it is just as *detectable*, confirming

<sup>1</sup>The small numeric difference for wall@4 — mode-firing rarity 0.1430 (3000 rollouts, Table 1) vs. reveal-rarity 0.1385 (2000 rollouts, Table 5) — is resampling noise on the same event, not a different event.

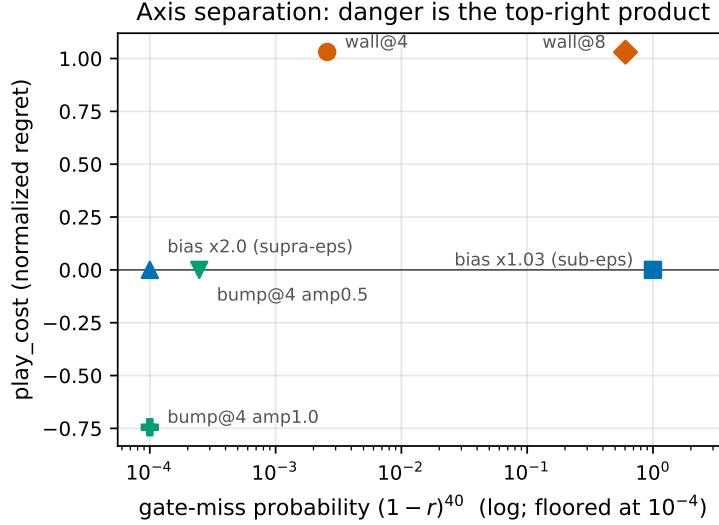


Figure 3: Table 5 as the danger quadrant: gate-miss probability (log) vs play\_cost. Only the rare hard mode reaches the top right; danger is the product of the coordinates.

Proposition 4 is about error geometry, not about hiding from the gate) yet play\_cost 0.000 at amplitude 0.5. At amplitude 1.0 play\_cost turns *negative* ( $-0.745$ ): the truth planner, seeing the slowdown near its horizon edge, is over-pessimistic and often settles for the small plateau, while the bump-blind planner pushes through and wins. A smooth localized omission produces planner-side timing effects of ambiguous sign; only the hard mode produces the one-way exploitation geometry (pinned, forever, below random).

**Is  $\varepsilon = 0.01$  a special setting, or is the axis separation itself  $\varepsilon$ -invariant?** A sweep over  $\varepsilon \in \{10^{-9}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 0.1, 0.3\}$  (`scripts/continuous_eps_sweep.py`; full table in `docs/EXPERIMENTS.md`) answers:  $\varepsilon$ -invariant. Mode-arm reveal-rarity is flat across the entire deployment-realistic range — on the cart, `wall@8` is bit-identically flat through the whole grid including  $\varepsilon = 0.3$ , and `wall@4` only dips slightly (never widens) at the top of the grid. The pervasive bias arms switch sharply at their own error scale on both instruments instead.

| $\varepsilon$ | wall@8 rarity | bias $\times 1.03$ rarity | bias $\times 2.0$ rarity |
|---------------|---------------|---------------------------|--------------------------|
| $10^{-6}$     | 0.0125        | 1.0000                    | 1.0000                   |
| $10^{-2}$     | 0.0125        | 0.0000                    | 1.0000                   |
| 0.1           | 0.0125        | 0.0000                    | 0.0040                   |
| 0.3           | 0.0125        | 0.0000                    | 0.0040                   |

Table 6: Reveal-rarity vs.  $\varepsilon$  (cart). The mode arm is flat across the whole grid; the pervasive arms switch at their own error scale.

The pendulum replicates both halves: its mode arms are flat too (only a slight dip at the top of the grid — `stop@1.0` rarity 0.1410 at  $\varepsilon \leq 3 \times 10^{-2}$ , dipping to 0.1400 at  $\varepsilon = 0.1$  and 0.1240 at  $\varepsilon = 0.3$ ), and its bias arms switch at the same error-scale boundaries as the cart’s. `pass@40`  $\approx (1 - r)^{40}$  continues to hold for the mode arms at every  $\varepsilon$  in the grid, on both instruments (honesty note: at `wall@8` the closed-form 0.6046 sits marginally below the empirical 300-gate Wilson 95% CI lower bound of 0.6115 — sampling noise at that gate count, reported plainly rather than as exact agreement). `play_cost` is not re-measured across  $\varepsilon$ : the model under test does not depend

on the gate’s tolerance, so play behavior is  $\varepsilon$ -independent by construction — the sweep varies only what the gate can see. The gate’s  $\varepsilon$  is a pervasive-error dial, not a mode-detection dial: tightening it cannot catch the hard mode, and loosening it does not widen the hole.

**The 4D bi-modal instrument replicates the flat mode arm, once per mode.** On PatchField2D the  $\varepsilon$ -sweep (`scripts/continuous_eps_sweep_patch2d.py`) gives *exactly* flat mode-arm reveal-rarity across the entire grid: patches-omitted 0.147, patch-1-only 0.142, patch-2-only 0.005 — each constant at every  $\varepsilon$ . The per-mode arms recover the per-mode rarities of the mechanism sweep (Section 4.2), so the tolerance axis is orthogonal to the mode hole on the bi-modal instrument too, and separately for each mode.

## 6 Mitigation: the exploitation is planner-mediated

The exploitation measured in Sections 4 and 5 is planner-mediated, not model-mediated, and a planner-side fix collapses it without touching the model or the gate — this does **not** contradict the danger law (the gate still certifies a wrong model; Proposition 2 is untouched). Like those sections, this one uses the hand-written instruments and blind models, not LLM synthesis. Distrust-region replanning (`src/cwm/continuous/mitigation.py`, strictly additive) compares the model’s prediction against the observed transition after every real step ( $\text{tol} = 10^{-6}$ ); a disagreement records the *position of the model’s refuted prediction* — not the pre-state — as a one-sided fence, since false predictions always lie on or beyond the mode boundary. While scoring a candidate rollout, the first imagined step whose position interval crosses a fence’s  $\varepsilon$ -band ( $\varepsilon = 0.25$  cart,  $\varepsilon = 0.1$  pendulum) truncates the rollout — reward kept, everything downstream dropped — which makes the fence leap-proof at any imagined speed; candidates are then ranked by (truncated return, distance to the nearest fence), which structurally prefers the real side. With zero violations this is bit-identical to plain MPC by construction (tested bitwise) — mitigation costs nothing when the model is right. Only this design is undodgeable: three earlier attempts (pre-state flee-balls; full-state point fences, dodged by the planner probing crossing velocities) were each defeated by the argmax planner acting as an adversary against the fence. The collapse is scoped to **hard-boundary hybrid modes**: the one-sided fence works precisely because a refuted prediction lies on or beyond the mode boundary, so fencing its far side cannot cut off any real trajectory. This structural fact is what these wall/stop instruments supply; less structured failure modes (soft or moving boundaries, errors not confined to a hard stop) do not obviously admit such a fence and are untested — this is a hard-boundary mitigation, not yet a general planner-side one.

The collapse is large at every knob: `pc_blind` stays pinned at  $\approx 0.94$ – $1.03$  everywhere (established in Section 4, Tables 1 and 2), while `pc_mit` never exceeds 0.81. Exactly one violation suffices to fence the mode on *every* one of the 11 rows — the mitigated planner must touch the mode once, which is identifiability operationalized: you cannot avoid what you have never seen. The residual `pc_mit` is the honest cost of that unavoidable first contact, and it grows with the lure distance, read off the first-contact step: cart  $0.290 \rightarrow 0.806$  as first contact goes  $11.6 \rightarrow 28.7$  of 80 steps (knob  $2 \rightarrow 10$ ); pendulum  $0.113 \rightarrow 0.212$  as first contact goes  $7.0 \rightarrow 13.0$  (knob  $0.8 \rightarrow 2.0$ ). The cart knob = 10 row is the least favorable in the sweep (`pc_mit` 0.806, close to blind’s 0.977) because the transient consumes most of the horizon — but the blind planner stays pinned *forever* there ( $J$  0.94 of  $J_{\text{truth}}$  17.77) while the mitigated planner escapes and recovers most of the horizon ( $J$  3.88): the normalized cost looks modest, the actual return does not.

| knob                                       | pc_blind | pc_mit | first-contact step |
|--------------------------------------------|----------|--------|--------------------|
| <i>cart</i> ( $x_{\text{wall}}$ )          |          |        |                    |
| 2                                          | 1.031    | 0.290  | 11.6               |
| 4                                          | 1.031    | 0.446  | 16.9               |
| 6                                          | 1.031    | 0.578  | 21.3               |
| 8                                          | 1.030    | 0.699  | 25.1               |
| 10                                         | 0.977    | 0.806  | 28.7               |
| <i>pendulum</i> ( $\theta_{\text{stop}}$ ) |          |        |                    |
| 0.8                                        | 1.002    | 0.113  | 7.0                |
| 1.0                                        | 1.002    | 0.129  | 8.1                |
| 1.2                                        | 1.000    | 0.143  | 9.0                |
| 1.4                                        | 0.997    | 0.160  | 10.0               |
| 1.6                                        | 0.990    | 0.177  | 11.0               |
| 2.0                                        | 0.942    | 0.212  | 13.0               |

Table 7: Mitigation sweep (20 episodes/knob, both instruments). `pc_blind` / `pc_mit` = `play_cost` of the blind planner under plain MPC / under distrust-region mitigation. Full 11-row table with contact rates and violation counts in `docs/EXPERIMENTS.md`.

### 6.1 The 2D mitigation: a partial collapse that decays with mode distance

The distrust-region fence generalizes to `PatchField2D` (`scripts/continuous_mitigation_patch2d.py`): fences are the 2D positions of refuted predictions, which lie *inside* an unreachable patch (the stay-at-previous-position clamp forces a refuted prediction strictly inside the disc, so fencing it cannot cut off any real trajectory), and a candidate rollout truncates when an imagined step *segment* passes within  $\varepsilon$  of a fence (segment-to-point distance, leap-proof); the 1D code path is preserved bit-identically (tested char-identical). But the collapse is now **partial and decays with patch distance** (Table 8): `pc_blind`  $\approx$  1.006 falls to `pc_mit` 0.257/0.541/0.862 at knobs (2,6)/(3,7)/(4,8), with mean violations 1.05/2.65/4.25 (against the 1D instruments’ clean 1.0) and first contact at step 8.5/12.35/15.25. The 1D single-violation sufficiency becomes a **boundary-mapping transient**: the planner rounds one fence disc and re-contacts the patch edge elsewhere, accreting fences along the probed arc, so more of the horizon is spent mapping the curved boundary as the patch recedes. At (4,8) the transient nearly defeats the mitigation (`pc_mit` 0.862 against `pc_blind` 1.006). This is the honest scope of a *hard-boundary* mitigation on a curved 2D mode — the one-sided fence still works, but a circular boundary costs a multi-contact mapping transient where a 1D clamp costs exactly one contact — not a failure of the fence.

| $(k_1, k_2)$ | pc_blind | pc_mit | mean viol. | first contact |
|--------------|----------|--------|------------|---------------|
| (2,6)        | 1.006    | 0.257  | 1.05       | 8.50          |
| (3,7)        | 1.006    | 0.541  | 2.65       | 12.35         |
| (4,8)        | 1.006    | 0.862  | 4.25       | 15.25         |

Table 8: `PatchField2D` mitigation sweep (paired seeds). The collapse is partial and decays with patch distance: mean violations grow  $1.05 \rightarrow 4.25$  (a boundary-mapping transient, versus the 1D instruments’ single violation), and `pc_mit` approaches `pc_blind` at the farthest knob.



## 7 LLM synthesis: the danger collapses to pure identifiability

Real-LLM arms (`scripts/continuous_danger_synthesis.py`; Azure GPT-5.x mini and large<sup>2</sup>;  $N = 40$  training rollouts which double as the gate, as in the companion paper’s sweep;  $\varepsilon = 10^{-9}$  pinned-integrator gate; **20 seeds/cell on the headline**  $x_{\text{wall}} = 8$  cell, **both sizes** (the companion paper’s standard); 6 MPC play episodes/seed; per-seed JSON with the synthesized code versioned in the repository). The contract pins the integrator (Section 2.1’s equations, stated in the spec text, constants generated from the environment instance so they cannot drift); the *full* arm includes the wall clause, the *incomplete* arm omits it. Crucially, each seed logs whether the wall fired in its training sample — the identifiability event that the companion paper could not condition on post hoc.

**One design fact governs every pooled count in this paper.** The gate sample of seed index  $i$  is drawn with rollout seed  $10^4(i+1)$  *independently of which model is being run* (`scripts/continuous_danger_synthesis.py`), so at every instrument and knob the mini and large arms are synthesized from the *same* samples. Pooling the two sizes therefore doubles the number of *synthesis draws*, not the number of independent gate samples: a pooled  $k/2n$  is  $n$  shared samples with two model draws each. Two consequences we hold to throughout. (i) Any empirical check of the gate-miss rate  $(1 - r)^N$  is a per-size statement — the mode-absent/present split is a property of the sample, so it is *identical* across sizes by construction and cannot be pooled at all. (ii) For the conditional outcomes (blind-and-exploited, repaired), pooling across sizes is meaningful because the synthesis draw is independent, but a Wilson interval over the pooled count assumes independence the samples do not have; we therefore quote the per-size bound as the conservative one and mark pooled bounds as such wherever they appear.

**Full arm (both sizes, 40 seeds): gate 1.000 in 0 refinement iterations, wall probes exact, play at truth parity — every seed.** The pinned-integrator premise holds with a real LLM: correct synthesis is float-exact through the sandbox, so  $\varepsilon = 10^{-9}$  costs nothing and the tolerance axis is fully disarmed. As in the discrete setting, given the rule, the model translates it perfectly.

**Incomplete arm.** Three-way structure:

1. **Wall absent from the sample** (the  $(1 - r)^N$  event; 10/20 seeds at  $x_{\text{wall}} = 8$  for each size, consistent with  $(1 - 0.0125)^{40} \approx 0.60$ ): every such seed — **20/20 across mini and large** — passed the gate at 1.000, fully wall-blind on the probes, and was exploited at play: pinned at the wall, contact rate 1.0, **play\_cost 0.999** (Wilson 95% lower bound 0.72 per size on 10 samples; the pooled 20/20 would give 0.84 but its 20 trials are 10 shared samples  $\times$  2 model draws, so we rely on the per-size bound). This is the discrete headline, synthesized end-to-end in a continuous CWM: a verified, almost-everywhere-exact model that performs worse than random.
2. **Wall present, repaired:** the LLM does *not* stay blind. It reads the failing transitions and writes the true global rule — `if x2 >= 8.0: return [8.0, 0.0]` (or the equivalent

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<sup>2</sup>**Exact models.** “large” is the Azure OpenAI deployment `gpt-5.4` and “mini” is `gpt-5.4-mini`, both at API version 2025-04-01-preview; the cross-family arms are `Qwen/Qwen3-Coder-30B-A3B-Instruct` (Hugging Face Inference Providers router) and Claude Sonnet (agent-relayed, Section 7). Every result JSON under `results/` records its own `model` field, so each number is attributable to a named deployment; the runs are dated 2026-07-07 to 2026-07-20 in the repository history. We write “GPT-5.x” in prose where the claim holds for both sizes.

if  $x_2 > 8.0$ :  $x_2 = 8.0$ ) — not a curve fit. At 20 seeds both sizes repaired **every** wall-present seed: **large 10/10 in 0–1 iterations** (two from the synthesis examples alone), **mini 10/10 in 0–5 iterations**. This is the divergence from the discrete setting, where rules demonstrated by example transitions were persistently not learned (translation-not-inference). A numerically-manifested discontinuity is learnable from data in a way a symbolic game rule was not.

3. **Wall present, not repaired:** at 20 seeds on the headline cell GPT-5.x produced **no stalls**, but stalls are real where they occur (the 5-seed  $x_{\text{wall}} = 4$  cell; the Qwen cross-family arm below). They are not near-misses of the rule but **superstitious local patches**: e.g. if  $\text{abs}(x_2 - 8.0) \leq 0.15$  and  $\text{abs}(v_2) \leq 1.1$ :  $x_2 = 8.0$ , or Qwen’s if  $x_2 \geq 8.0$  and  $v_2 \leq 0.0$ : ... — clamps fitted to the *observed manifestation* of the mode (low-speed, near-wall contacts), which mispredict other approaches to the wall. **The gate rejected every one of them** (gate 0.49–0.999, never 1.000).

**Cross-family spot-checks (two families).** *Qwen* (HF router, Qwen/Qwen3-Coder-30B-A3B-Instruct, 3 seeds): the full arm is clean (3/3 gate 1.000, blind 0.0), so the pinned-integrator premise is not GPT-specific; on the incomplete arm the identifiability branch reproduces (the 1/3 wall-absent seed is gate-1.000 wall-blind and exploited, play\_cost 0.999), but Qwen *repaired neither* of its two wall-present seeds (gate 0.999 and 0.491, both superstitious patches the gate refused). *Claude* (Sonnet, agent-relayed via the companion paper’s protocol on `scripts/continuous_claude_step.py` — verbatim pipeline messages relayed to fresh, context-free instances per message, same  $\varepsilon = 10^{-9}$  gate and MPC play as the API arms; seeds 10000/20000/30000 plus one full control per instrument; `results/continuous_claude_relay.json`): both full controls are clean (2/2 gate 1.000, blind 0.0, play\_cost 0.0), and both mode-absent seeds are certified fully blind and exploited (cart play\_cost 0.999, pendulum 0.995) — the identifiability event fires family-independently, as Proposition 2 requires. On repair Claude is neither GPT-5.x nor Qwen: cart seed 20000 repaired the exact one-sided rule in 1 iteration, and cart seed 30000 repaired at iteration 5 after a period-2 oscillation (symmetric  $\pm 8$  walls  $\rightarrow$  both removed  $\rightarrow$  symmetric again  $\rightarrow$  removed  $\rightarrow$  one-sided correct), but pendulum seed 20000 was certified in 1 iteration carrying an invented mode (next paragraph) and pendulum seed 30000 stalled at 5 iterations (gate 0.9972), oscillating between the symmetric-stop and no-stop artifacts without ever finding the one-sided rule. So across three families the mode-absent blind-and-exploited event fired for every one (it is a property of the sample; Proposition 2), while repair-from-data is model-dependent *in mechanism, not merely in rate*: GPT-5.x repairs every revealed mode exactly (62/62 pendulum, 20/20 cart), Qwen repairs none (superstitious local patches), and Claude repairs most but through a *symmetry prior* — it generalizes one-sided boundary evidence into a symmetric pair of boundaries.

**A fourth artifact class: certified with an invented mode unfalsifiable by its gate sample.** Claude’s symmetry prior produces an outcome beyond the correct/blind/superstitious-patch trichotomy. Where the training sample covers the invented side, the gate refutes the phantom and the memoryless refine loop oscillates (cart seed 30000’s period-2 cycle above; pendulum seed 30000’s stall at gate 0.9972). But where the sample is *silent* on the invented side, the phantom is unfalsifiable: pendulum seed 20000 was certified at gate 1.000 in a single iteration while carrying a phantom symmetric stop at  $\theta = -1.4$  that this seed’s rollouts never reach (if `th2 < -th_max`: `th2 = -th_max`; `om2 = 0.0`) — a verified artifact, exact on every sampled transition, that nonetheless encodes a hard stop which does not exist. This is a clean natural experiment: the

*same* symmetric artifact was refuted at pendulum seed 30000 and certified at seed 20000, the only difference being whether the sample covered  $\theta < -1.4$ . It is Proposition 2’s prior caveat measured directly — on inputs the sample never touches, the artifact’s content comes from the model’s prior, and the gate cannot police it. Note the classification did not flag it: the `mode_blindness` probe scored 0.0 (correct) because it probes only the true +1.4 mode’s region; code inspection, not the probe, caught the invented mode (Section 10).

**Honesty notes (Claude arm).** (i) Agent-relayed means an agent scaffold over a subscription transport, not an API, though the relayed messages are byte-identical to the pipeline’s; in the two multi-iteration cells a handful of refinement replies prefixed a one-line explanation before the code block (two distinct sentences, repeated across the oscillation) despite the output-only-code instruction — recorded, and the code block still parsed; no relay was refused this time (contrast the discrete probe). (ii) The refine loop is memoryless in the API arms too — `refine_continuous` sends a single user message per iteration (`src/cwm/continuous/contract.py`) — so the oscillation is protocol behavior, not a relay artifact. (iii)  $n$  is small: 3 seeds plus one control per instrument, one alternate family.

The branches compose into the paper’s central claim. With the mode in the data, the synthesize–refine–gate loop behaved soundly *on every transition the sample covered*: it recovers the exact mode or refuses the artifact (sample-covered exactness is the acceptance criterion itself; the empirical, code-inspected regularity — not a theorem — is that every accepted artifact’s off-sample content was also the true mode, except Claude’s phantom pendulum stop, which errs only where its sample is silent: the Proposition 2 prior caveat, not a gate failure). With the mode absent, no loop can help (Proposition 2: the sample carries no evidence for the mode — though a prior or the specification still could supply it), and the acceptance of a blind artifact is not a failure of the LLM, the refinement, or the gate implementation — it is the sampling event whose probability is exactly  $(1 - r)^N$ . The discrete danger law had a provable core plus an empirical residual (the rule not learned even when shown); on the continuous 1D instruments the residual *can* vanish — it does for GPT-5.x, which repairs every revealed clamp — so **the law becomes the entire failure surface** for a capable-enough synthesizer, and shrinks toward it even for a weaker one (this is geometry-scoped: Section 7.1 shows the residual returns on a 2D circular mode, where even GPT-5.x does not repair). The actionable consequence sharpens correspondingly: in this regime, spec completeness and gate-sample coverage are not two independent worries — coverage *is* the dominant worry, because the synthesis loop repairs what the sample reveals.

(Scope: 20 seeds/cell on the headline cell, both GPT-5.x sizes; the wall-absent conditional is 20/20 across sizes and consistent across three independent runs — the 5-seed first run, the 20-seed tightened run, and the 3-seed Qwen run; the wall-present repair rate is 20/20 for GPT-5.x on this cell, 0/2 for Qwen, and mechanism-dependent for Claude (exact on both cart seeds, one via a period-2 oscillation) — model-dependent, small- $n$  on the cross-family arms. LLM synthesis is stochastic across calls; the three-way structure and the identifiability conditional are stable run-to-run, per-seed iteration counts are not.)

**Second-instrument robustness (pendulum-with-stop, Section 4.1, 20 seeds/cell, both sizes).** The synthesis arm is not cart-only. Running the identical pipeline on the nonlinear pendulum — headline  $\theta_{\text{stop}} = 1.4$  (rarity 0.019) and caught  $\theta_{\text{stop}} = 1.0$  (rarity 0.128; “caught” labels a cell whose mode is common enough that the gate sample essentially always contains it, so the identifiability event essentially never fires) — reproduces every branch, including a 3-seed Qwen cross-family spot-check at the headline knob (Table 9).

| cell (20 seeds each)                        | full  | absent $\rightarrow$ blind & exploited | present $\rightarrow$ repaired (stalled) |
|---------------------------------------------|-------|----------------------------------------|------------------------------------------|
| mini $\theta_{\text{stop}} = 1.4$           | 20/20 | 9 $\rightarrow$ 9 (pc 0.995)           | 11 $\rightarrow$ 11 (0)                  |
| large $\theta_{\text{stop}} = 1.4$          | 20/20 | 9 $\rightarrow$ 9 (pc 0.995)           | 11 $\rightarrow$ 11 (0)                  |
| mini $\theta_{\text{stop}} = 1.0$           | 20/20 | 0 $\rightarrow$ —                      | 20 $\rightarrow$ 20 (0)                  |
| large $\theta_{\text{stop}} = 1.0$          | 20/20 | 0 $\rightarrow$ —                      | 20 $\rightarrow$ 20 (0)                  |
| Qwen $\theta_{\text{stop}} = 1.4$ (3 seeds) | 3/3   | 1 $\rightarrow$ 1 (pc 0.995)           | 2 $\rightarrow$ 0 (2 stalled @0.9997)    |

Table 9: Pendulum synthesis cells (20 seeds/cell, both GPT-5.x sizes; 3-seed Qwen spot-check at  $\theta_{\text{stop}} = 1.4$ ).  $k \rightarrow m$  = of the  $k$  seeds in that branch,  $m$  had the stated outcome (blind & exploited, resp. repaired); the parenthetical count is stalled seeds; pc = play\_cost.

Pooled across both knobs and both sizes (Table 9), every mode-absent occurrence was blind and exploited at play\_cost 0.995 (per-size headline 9/9, Wilson 95% lower bound 0.701 — the bound we rely on; the pooled 18/18 would give 0.824 but its two sizes share the same 9 samples, Section 7) — the same fixed-point exploitation as the cart’s play\_cost  $\approx 1$  — and GPT-5.x repaired **62/62** mode-present seeds to the exact angular clamp (if `th2 >= theta_stop: return [theta_stop, 0.0]`), 0 stalls (Wilson 95% lower bound 0.942, pooled over two knobs and two sizes on 31 shared samples; the per-size, per-knob cells are 11/11 and 20/20). Qwen reproduces the mode-absent blind-exploited event but stalls on both its mode-present seeds, the same superstitious-patch signature as on the cart; Claude’s agent-relayed spot-check on this instrument produced the phantom-mode artifact discussed in Section 7 (a certified stop at  $\theta = -1.4$  its sample never reaches, plus a stall at the other mode-present seed). Repair is model-dependent; identifiability, being a property of the sample, is not — on this instrument too. The mechanism was already validated on two instruments (Section 4.1); the synthesis result now is as well: a nonlinear plant with an angular, not positional, hard stop reproduces the same danger law and the same repair capability, so the repair finding is not a cart artifact.

## 7.1 The repair finding inverts on a 2D circular mode

Everything above — and the (b)-residual-vanishes claim of the Introduction — was measured on *one-dimensional* hard boundaries: a position clamp (cart) and an angular clamp (pendulum), on which GPT-5.x repaired every revealed mode (**82/82** pooled — 41 distinct samples with two synthesis draws each, Section 7). PatchField2D asks whether that repair survives a genuinely 2D mode — a circular boundary  $(x' - c_x)^2 + (y' - c_y)^2 \leq R^2$  — and whether an artifact can repair one patch while staying blind to the other (**partial repair**, which the design predicted and per-mode blindness makes measurable). We ran the identical pipeline (Azure GPT-5.x mini and large, 20 seeds/cell,  $N = 40$ ,  $\varepsilon = 10^{-9}$ ) on two bi-knob cells,  $k = (3, 7)$  and  $k = (5, 9)$ , classifying each seed by which modes its training sample contained {miss both, see one miss the other, see both}. The full arm is clean (20/20 per cell, both sizes).

**We predicted partial repair; we measured none.** Across the **66** see-one-miss-the-other seeds (64 see- $P_1$ -miss- $P_2$  plus 2 miss- $P_1$ -see- $P_2$ ), **zero** produced a partial-repair certificate — not one artifact repaired the seen patch and stayed blind to the unseen one while passing the gate. More strongly, of the **76** incomplete-arm seeds whose sample contained *at least one* mode, **0/76** recovered the circular disc rule at all.

**The mechanism: translation succeeds, induction of the boundary collapses.** A 76-artifact code inspection locates the failure precisely. *Plant translation succeeds*:  $\approx 74/76$  artifacts

reproduce the exact 4D semi-implicit integrator and the reward. What collapses is *induction of the 2D circular boundary*. The dominant modal failure (38/76) is **dimensional reduction**: the disc is modeled as a *half-plane* — a 1D CartWall-style clamp at the right location but the wrong shape, e.g. substituting `if x2 > 4.0:` ... for a circle — learning the freeze mechanic but not its circular trigger. The remaining classes: 20/76 pure-blind (no patch logic), 9/76 superstitious local patches, and 9/76 disc-form attempts, none of them correct. The  $\varepsilon$ -exactness alternative — that correct-form discs merely failed the  $10^{-9}$  arithmetic gate — is **falsified**: no correct-form disc failed on arithmetic; the discs that failed were the wrong *shape*. A *behavioral* audit confirms this hand inspection on every claim (`scripts/patch2d.artifact.audit.py`: probe each artifact’s `step()` on a state grid, classify the shape of its deviation-from-integrator set): 39/76 half-plane-form, 13 behaviorally blind, 8 disc-form, 7 measure-zero textual traps, 4 point, 3 square-form, 2 bounded-other; integrator arithmetic exact in 74/76; and **no artifact covers even its seen patch** — the missing partial repairs were never attempted, not merely refused by the gate. The two class counts differ by exactly one artifact (39 vs 38 half-plane, 8 vs 9 disc-form) and the difference is a *syntax-versus-behavior* reading, not a disagreement about any artifact’s correctness: two artifacts (mini  $k = (3, 7)$ , seeds 130000 and 150000) write a *radial* predicate anchored at the reward lodes rather than at a patch — `if hypot(x-(-6),y) <= 2 or hypot(x-12,y) <= 2` in one, a reward-threshold freeze in the other — so hand inspection reads the disc in the source while the audit reads the deviation set, which is unbounded inside the probed box. Both readings agree that neither artifact encodes a patch; only the label moves.

**Ablation 1: richer prompting and  $3\times$  budget do not restore repair — they change the failure class.** The natural confound is that the prompt was too poor and the budget too small. The strongest joint treatment (120 examples, 40 failure lines, describe-the-region-first guidance with an explicit “the region need not be a 1D threshold” de-bias, and 15 refine iterations; identical samples to the original run) yields **0/40 repair** at  $k = (3, 7)$ , both sizes. The guidance *works as prompt engineering*: the half-plane reduction disappears (0/40, versus 21/40 in the matching base cells) and the artifacts now write bounded 2D regions — rotated ellipses ( $\sim 15/40$ ), rectangles ( $\sim 10/40$ ), unions of micro-discs ( $\sim 5/40$ ) — but none is the true disc. The new failure class is *evidence-hull fitting* with two compounding errors: they fit the hull of the *observed* freeze positions (the pre-freeze crescent hugging the disc’s reachable west boundary from outside: fitted centers  $x \approx 2.3$ , outside a patch whose west edge is at  $x = 2$ ), and 36/40 condition on the current position rather than the landing  $(x_2, y_2)$  — the causal variable of the rule. The sample itself then refuses every wrong shape.

**Ablation 2: a zero-curvature square falsifies the curvature reading.** If the failure were about *curved* boundaries, an axis-aligned square patch (a Chebyshev ball,  $\max(|x - c_x|, |y - c_y|) \leq R$  — a `max/abs` predicate, no quadratic) should be repairable. It is not: with the same pipeline at  $k = (3, 7)$  the full arm is clean (20/20 at zero refine iterations, both sizes — translating the square clause is as easy as the disc’s) and the incomplete arm gives **0/40 repair** (every sample mode-containing; best gate 0.9962). The classes mirror the disc’s, *reflected*: the dominant failure is again dimensional reduction (the square’s west edge as a 1D threshold, `if x2 >= 2.0`), and several artifacts commit the *inverse* error — writing *discs* on square evidence (radial `hypot` guards at the edge midpoint, micro-disc unions), plus reward-anchored superstitions (freeze zones invented at the lodes). Zero artifacts write the true box or its `max/abs` form. Pooled across the three campaigns: **0/156** mode-containing seeds recover the 2D region — which, in the shared-sample accounting of Section 7, is 78 distinct gate samples each given two independent synthesis draws (38 disc +



20 square + 20 guided). For a *negative* result the pooling is the mild direction: each draw is an independent opportunity to repair, and none took it.

**Ablation 3: a second family fails in the same classes.** The three campaigns above are GPT-5.x only, so the collapse could be one family’s idiosyncrasy. A Claude arm on the headline cell says otherwise (agent-relayed as in Section 7; Sonnet,  $k = (3, 7)$ , three mode-containing seeds — two see- $P_1$ -miss- $P_2$ , one seeing both — plus one full control, same  $N = 40$  sample, same  $\varepsilon = 10^{-9}$  gate, same five-iteration memoryless refine loop). The full control is clean at iteration 0 (gate 1.000, both discs written on the *landing* position, blindness 0.0, play\_cost 0.0): translation is not the problem here either. No incomplete seed repaired (0/3), and the *trajectory* carries more information than the outcome: each seed spends its five iterations in a **period-2 cycle** between the pure-blind artifact and a wrong template, returning to its iteration-0 gate value exactly on every even iteration (seed 10000: 0.9934 blind  $\rightarrow$  0.9591 disc-on-*current*-position  $\rightarrow$  0.9934  $\rightarrow$  0.9284 half-plane  $\rightarrow$  0.9934  $\rightarrow$  0.7044 reward-threshold; seed 20000: blind 0.9962  $\leftrightarrow$  half-plane 0.9406, three times over; seed 30000: 0.9966  $\rightarrow$  0.6750 reward-zone  $\rightarrow$  0.9966  $\rightarrow$  0.8406 half-plane  $\rightarrow$  0.9966  $\rightarrow$  0.5328 *y*-band). The template set is the one the GPT-5.x campaigns exhibited — 1D threshold, radial disc on the *wrong* (current, not landing) variable, reward-landmark zone, axis-aligned band — and **no incomplete iteration in any seed wrote a disc on the landing position**, the form this same model writes immediately when the contract states it. The *constants* replicate Ablation 1 quantitatively: every half-plane sits at  $x = 2.0$ , the disc’s west edge — “the right location, the wrong shape” — and the one radial attempt fits center (2.328,  $-0.135$ ) with radius 0.824, i.e. the hull of the observed freeze crescent centered *outside* a patch whose west edge is at  $x = 2$ , the same  $x \approx 2.3$  signature the guided GPT-5.x artifacts produced. Per-iteration gate accuracies and rule classes are re-derived from the versioned transcripts by `scripts/claude_relay_ledger.py`. Two honest notes. (i)  $n = 3$  seeds, one alternate family on the *incomplete* arm: this closes the “GPT-only” confound, it does not sweep models. A third family (Qwen) was launched on this cell and **aborted after its three full-arm cells** when that provider’s inference credits ran out, so it contributes a translation control and nothing else — but that control is clean and worth recording: 3/3 gate 1.000 at zero refinement iterations, both discs written exactly on the landing position, per-patch blindness 0.0. Across all three families, then, *stating* the region rule is easy and *inducing* it from data is what fails; the induction evidence on this instrument is GPT-5.x’s 156 seeds plus Claude’s 3. (ii) The relay caveats of Section 7 apply, plus one specific to this arm: the relayed instances were instructed not to use tools or read files, so each artifact is a function of the pipeline message alone (the ground truth lives in this repository, so an instance reading it could have copied the true centers and radius instead of inducing them). The wrapper text is recorded verbatim in `results/claude_relay_transcripts/patch2d_k3_7_RELAY_FRAMING.txt`, together with the post-hoc leak check: no incomplete artifact wrote the true disc form at all, and none names the unseen far patch — the signature repository access would have left.

**The gate is all-or-nothing, and stays sound.** Certification occurred *iff* the sample missed *all* modes: the only certified incomplete artifacts were the **4/80** miss-both seeds (2 per size at  $k = (5, 9)$ ), blind on both patches and exploited at play\_cost 1.095, contact rate 1.0 — the doubly-blind fixed point, at the joint-miss event of probability  $(1 - r_1)^N(1 - r_2)^N$ . Every see-one-miss-the-other artifact was *rejected*, not certified: the all-or-nothing gate refused every partial artifact rather than certifying a half-repair. Certification therefore stayed sound in the strict sense — no wrong artifact passed on a transition its sample covered. Two cautions on reading the failing gates: a high near-miss score (e.g. 0.9997) reflects the *rarity of a mode contact* in a short rollout,



*not* a near-repair, so more refinement iterations would not populate a partial-repair branch that does not exist; and the mini/large partitions are seed-identical by construction (shared seeds — the design fact of Section 7, which holds at every instrument, not only here), so the two sizes are not independent replications.

**Repair-from-data is geometry-dependent — and the axis is a template prior, not curvature.** The arc is the paper’s own: on 1D clamps the discrete paper’s (b)-residual *vanished* (GPT-5.x repaired 82/82); on the 2D modes it *returns* (0/156). Repairing an omitted hybrid mode from data is therefore **geometry-dependent**, and the two ablations locate the axis: it is *not* boundary curvature (the flat square fails identically, with the errors reflected) and *not* prompting or budget (the strongest treatment fails, differently). The mechanism the three campaigns agree on is a **template prior**: the synthesizer selects from a small library of low-descriptive-complexity region forms — 1D threshold, radial ball, reward-landmark zone, evidence hull — and keeps whatever locally fits, under-weighting the evidence’s actual geometry in *both* directions (flattening curves, curving flats). Dimensional reduction is one face of this prior; the square run exposes the other. Induction of a 2D generative boundary from one-sided contact evidence is exactly what the prior replaces, and the danger law’s empirical exhaustiveness — “the residual vanishes, so coverage is the whole game” (Section 7) — is correspondingly **geometry-scoped**: it holds for the 1D clamps and fails on 2D regions, where a capable synthesizer translates the plant exactly yet substitutes a template for the mode. What does *not* change is the provable core: identifiability and the gate-miss law hold verbatim, certification stayed sound, and the danger still lives exactly in the joint-miss event. The residue is that on 2D geometry the loop no longer repairs what the sample reveals — so the discrete paper’s exhaustiveness claim, sharpened to a boast on the 1D instruments, is scoped back by the very move (harder geometry) that was supposed to test it.

## 8 Smooth learners cannot localize: both halves, measured

If localization is representational, two things must be checkable on non-code learners trained on the *same data* (`scripts/continuous_smooth_probe.py`; the two most favorable smooth learners: closed-form linear least squares — off the wall, the dynamics are *exactly linear*, so this is the smooth best case — and a small tanh MLP, probe-grade):

| model      | trained on | off-mode err mean / max | probe err | gate $10^{-9}$ | gate $10^{-2}$ |
|------------|------------|-------------------------|-----------|----------------|----------------|
| linear-LSQ | wall-free  | 3.6e−15 / 1.7e−14       | 4.18      | <b>PASS</b>    | <b>PASS</b>    |
| linear-LSQ | wall-data  | 1.9e−03 / 1.2e−02       | 4.17      | fail           | fail           |
| MLP h=8    | wall-free  | 3.5e−03 / 5.0e−02       | 4.20      | fail           | fail           |
| MLP h=8    | wall-data  | 6.0e−03 / 4.9e−02       | 4.19      | fail           | fail           |

Table 10: Smooth learners on the synthesis samples ( $x_{\text{wall}} = 8$ ). “probe err” is the wall-region probe error ( $4.2 \approx$  predicting straight through the wall).

**Identifiability is learner-independent, live.** On the wall-free sample the linear model recovers the off-mode dynamics to  $10^{-15}$ , **passes the  $\varepsilon = 10^{-9}$  gate**, and is exactly as wall-blind as the synthesized blind code (probe error 4.18: it predicts straight through the wall). Proposition 2 instantiated on a second hypothesis class: in the gate-miss event, *any* accepted learner is blind. The  $(1 - r)^N$  hole is not an LLM property.

**With the mode in the data, code and smooth part ways.** Four contact rows out of 3200 tilt the linear fit by **twelve orders of magnitude** off-mode ( $1.7\text{e}−14 \rightarrow 1.2\text{e}−02$  max; Figure 4)

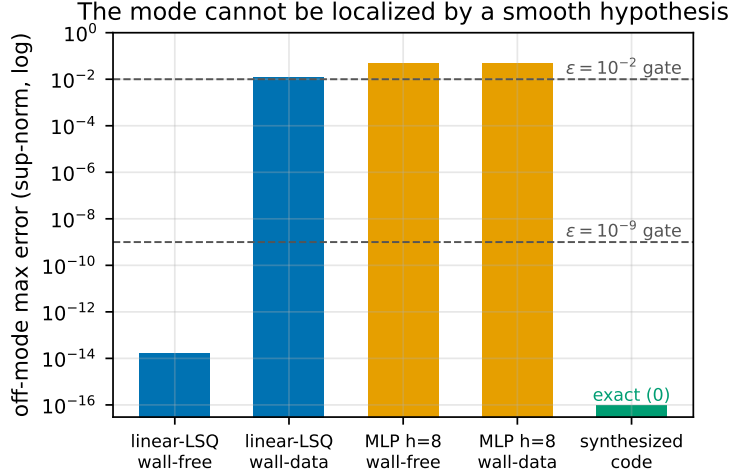


Figure 4: Off-mode max error (log) by model and training data. The wall-free linear fit sits at float noise and passes both gates (blind); four contact rows tilt it twelve orders of magnitude; synthesized code is exactly zero off-mode.

— it fails both gates *and still* has the mode wrong (probe 4.17). The smooth hypothesis cannot put the error on the mode; it leaks everywhere (Proposition 4’s geometry, observed as a least-squares tradeoff). The synthesized code, on the *same sample*, wrote the exact clamp and passed at float precision. The MLP combines both axes: a pervasive  $\sim 5e-3$  floor that no gate accepts, and a never-learned mode.

Consequence for the CWM paradigm: the paradigm *creates* in continuous control the localized failure class that the learned-model literature says does not dominate there — and, by the same representational fact, also creates the exact-repair capability that learned models lack. Code cuts both ways, and the gate’s sampling coverage decides which way.

## 9 Related work

*Objective mismatch and model exploitation in model-based RL* (Lambert et al., 2020; Janner et al., 2019; Hafner et al., 2020): prediction accuracy and control performance diverge for learned continuous models; planners exploit model errors. Our contribution to that conversation is the *verified-but-wrong localized* regime, which smooth learned models cannot even represent (Proposition 4, Table 10), plus a closed-form acceptance-failure law confirmed at gate scale. *Hybrid systems*: mode detection and identification of piecewise/hybrid dynamics is classical (Paoletti et al., 2007; Bemporad and Morari, 1999); our question is not identifying the mode but what a *sampling verifier certifies* when the mode is missed, and what a planner then does. *Code World Models* (Lehrach et al., 2026) and code-as-policy (Liang et al., 2023; Gao et al., 2023) supply the paradigm; the companion paper (Aguilar Martín, 2026) is the discrete study this paper extends — transferring its provable core, showing its empirical residual (translation-not-inference) does not transfer, and adding the representational theorem separating code from smooth hypotheses. *Property-based testing and rare events* (Claessen and Hughes, 2000; Rubinstein and Kroese, 2017): the gate is random testing of a program against an oracle;  $(1 - r)^N$  is the standard rare-input coverage gap, here with the planner as the adversary that finds it. *Falsification and rare-event testing of hybrid and cyber-physical systems* (Annpureddy et al., 2011; Corso et al., 2021): this literature searches for

inputs that drive a hybrid system to violate a specification; our gate is the passive dual — uniform random testing whose blind spot is precisely the rare mode a falsifier would hunt for, and the  $(1 - r)^N$  factor quantifies the coverage gap that random testing leaves. *Runtime monitoring and shielding for safe RL and control* (Alshiekh et al., 2018): a synthesized shield overrides unsafe actions using an a-priori specification; our distrust-region fence is instead deployment-time feedback, built from observed prediction failures rather than a given safety automaton, and it corrects a certified-but-wrong model rather than an unsafe policy. *Robust MPC and planning under model uncertainty* (Rawlings et al., 2017): robust and tube-based MPC guarantees performance against bounded model error; the failure here is orthogonal — the error is unbounded but localized to an omitted mode and is certified away by the gate, so robustness margins tuned to the off-mode error do not see it. *System identification of contact-rich and discontinuous dynamics* (Paoletti et al., 2007; Fazeli et al., 2017): estimating parameters and contact forces of rigid bodies undergoing frictional contact is a mature sysid problem; our repair-from-data result is the code analogue — given contact transitions in the sample, the synthesizer recovers the exact switching rule, whereas a smooth fit cannot localize it.

## 10 Limitations and Honest Assessment

**Three instruments; repair is geometry-scoped, mitigation decays with mode distance.** Cart-with-wall, pendulum-with-stop, and PatchField2D now span 1D and 4D state and one and two modes, so the earlier “one dimension, single stationary boundary” limitation no longer holds. The *mechanism* survived every extension — it is measure-theoretic — and so did strict gate soundness (the all-or-nothing 2D gate certified no partial artifact). The honest residue is now sharper and different. **Repair-from-data collapses on 2D region boundaries, curved or flat:** GPT-5.x repaired every 1D clamp (82/82) but recovered the 2D region rule in 0/156 mode-containing seeds pooled over the disc, the square ablation, and the guided/3 $\times$ -budget treatment (Section 7.1). The prompting/budget and curvature outs are now *measured negatives*; what could restore repair — two-sided (inside) evidence, active boundary probing, or richer evidence summaries — is the subject of ongoing follow-up work. The **2D mitigation** likewise generalizes but only partially, its collapse decaying with mode distance as a boundary-mapping transient that nearly defeats the fence at the farthest knob (Section 6.1). Contact-rich manipulation, moving boundaries, and 3+ modes remain future work.

**Two planner families, one fixed configuration each.** Random-shooting MPC occupies the high-query-reach/high-play-cost branch of Proposition 3; CEM occupies the low-query-reach/near-zero-play-cost branch (Section 4.3). This is stronger than a one-family result but not planner-universal: CEM was measured at one prototype-fixed setting, with no hyperparameter sweep, and its pendulum truth returns expose local optima. Its non-exploitation is reach-limited, not knowledgeable or safe — the same search that misses phantom reward can miss real distant reward. Gradient-based shooting and tree search (Coulom, 2006; Kocsis and Szepesvári, 2006) remain untested. Distrust-region replanning (Section 6) is a mitigation layered on MPC, not a third family; its hard-boundary guarantee and its behavior with other planners remain untested.

**Synthesis cells are modest, and the two sizes share their samples.** The gate sample is drawn from the seed index alone, so mini and large are synthesized from identical samples at every instrument and knob (Section 7): every pooled “both sizes” count is  $n$  distinct gate samples with two synthesis draws each, and the distinct-sample  $n$  is half the pooled one — 10 mode-absent samples

on the cart headline cell, 31 mode-present across the two pendulum knobs, 38 mode-containing on the PatchField2D disc cells, 20 on each ablation campaign. Per-size Wilson bounds are quoted for exactly this reason, and the  $(1-r)^N$  gate-miss check is never pooled (the mode-absent/present split is a property of the sample and is identical across sizes by construction). With that accounting: 20 seeds/cell on the headline  $x_{\text{wall}} = 8$  cart cell for both GPT-5.x sizes, plus a 3-seed Qwen cross-family spot-check; the pendulum arm adds two knobs (headline and caught) at the same 20 seeds/cell, both sizes, plus its own 3-seed Qwen spot-check. The wall/mode-absent conditional is 20/20 across cart sizes and three runs and 18/18 across pendulum sizes: per size that is 10/10 (Wilson lower bound 0.72) and 9/9 (0.70), the bounds we rely on, against pooled 0.84 and 0.824 whose trials share samples. The GPT-5.x repair rate is 20/20 on the cart headline cell and 62/62 pooled on the pendulum (31 distinct samples). The cross-family arms are small- $n$  (3 seeds plus one control per instrument per family) and cover two alternate families (Qwen, and Claude agent-relayed), not a sweep of models — and their coverage is uneven across instruments: both families were run on the two 1D instruments, but on PatchField2D only Claude has an incomplete arm — the Qwen run there aborted after its three full-arm cells when that provider’s inference credits were exhausted, so it is a translation control (3/3 exact) with no induction data, a gap of circumstance rather than of judgement; they show repair is model-dependent *in mechanism* — Qwen 0/2 mode-present (superstitious patches), Claude repairing most but via a symmetry prior that oscillated on two seeds and certified one phantom, unfalsifiable mode. The PatchField2D synthesis arm adds two bi-knob cells (20 seeds each, both GPT-5.x sizes) plus two ablation campaigns (region-guidance  $3\times$  budget; zero-curvature square — 20 seeds each, both sizes); its finding is a *negative* one — 0/76 repair of the circular mode, 0/66 partial repair, 0/156 pooled, plus 0/3 in a Claude cross-family arm on the same cell whose failure classes match (Section 7.1) — so repair-from-data is not only model-dependent but geometry-dependent, via a template prior.

**The mode-blindness probe covers only the true mode’s region.** By construction `mode.blindness (src/cwm/continuous/contract.py)` fires probes where the truth’s mode is active, so it certifies whether the *revealed* mode is encoded but is blind to a mode the model *invents* elsewhere: Claude’s phantom pendulum stop (Section 7) scored blindness 0.0 while carrying an extra, non-existent stop at  $\theta = -1.4$ . Detecting invented modes needs code inspection, or probes seeded outside the sampled region — the classification alone does not suffice, and our phantom-mode finding rests on reading the certified code, as the companion paper’s artifact-level analysis does.

**The MLP is a probe, not a baseline.** It substantiates the representational point at  $h=8$ /pure-Python scale; a tuned modern dynamics model would have a lower floor but the same structural inability to be bit-exact off a mode at  $\varepsilon = 10^{-9}$  (an argument, not yet a measurement, at scale).

**Proposition 4 bounds geometry, not probability.** The ball is metric; converting to gate-visitation probability needs the gate measure on that ball, which is instrument-specific. The empirical complement (Table 10’s twelve-orders tilt) carries the quantitative weight here.

**Coverage certificates do not transfer.** The companion paper’s enumeration-based guarantees have no continuous analogue in this paper; covering-number versions under Lipschitz assumptions are open.

**play\_cost > 1 is a normalization artifact** (blind < random), reported unclamped and explained; the headline claims never depend on the excess over 1.

## 11 Conclusion

The companion paper ended by diagnosing the gate failure as a reach-distribution shift and prescribing verification on the distribution the planner visits. This paper shows the diagnosis is not about discreteness. The gate-miss law, the identifiability argument, and the play-cost bound are measure-theoretic and survive the move to continuous state spaces intact; what the move changes is *where the localized failure can live* (hybrid mode boundaries — an omitted mode is a rare rule) and *who can express it* (programs, not smooth function classes: Proposition 4 and Table 10). On two minimal hybrid instruments the entire discrete phenomenology reproduces — threshold law, knob-invariant exploitation, the reach mechanism — with the gate’s acceptance rate closely tracking  $(1 - r)^N$  at gate scale (within sampling noise; Section 5).

The synthesis experiment then delivers this paper’s own finding: on the 1D instruments the continuous regime is the one where the danger law is *everything*. Given the mode in its sample, the LLM repairs it exactly — including from the synthesis examples alone — and when it fails to repair, the gate catches the superstitious patch; given the mode absent, every learner we tried (LLM, exact linear regression) is certified blind and the planner is exploited to below-random performance. That repair, though, is geometry-dependent: on a third, 4D bi-modal instrument with 2D modes it inverts — neither GPT-5.x size recovers the region rule from data (0/156, including a zero-curvature square and a guided 3×-budget treatment: a template prior over region forms, not curvature), and a second family fails in the same classes, and the danger law’s exhaustiveness is scoped back to the 1D clamps, though the gate stays sound (Section 7.1). The provable core is the gate-miss and identifiability propositions (Section 3): the mode-absent event is a hole no learner can close *from the sample* — a prior or the specification could still supply the mode — and its probability is in closed form. Sample-covered exactness is the acceptance criterion itself; that accepted artifacts were also right off-sample — with the single phantom-mode exception, wrong only where its sample is silent — is an empirical regularity of the models tested (GPT-5.x repairs; Qwen stalls and is refused; Claude repairs most through a symmetry prior), not a theorem. For practitioners the message is one clause sharper than the discrete one: in continuous code-world-model pipelines, *sample coverage of the mode boundaries is the whole game* — the synthesis stack will translate what it is told and repair what it is shown, and no stage of it can know about what the sample never touched.

## A Reproducibility

All code is at <https://github.com/JaviMaligno/code-world-models> (src/cwm/continuous/, scripts/continuous\_\*.py); the experiment log is docs/EXPERIMENTS.md (sections prefixed “PA-PER 2”) and per-seed artifacts, including every synthesized program, are versioned under results/. CPU results (Tables 1–10):

```
PYTHONPATH=src python scripts/continuous_reach.py
                        # danger-curve table (tab:danger; ~2.5 min)
PYTHONPATH=src python scripts/continuous_axes.py
                        # axis-separation table (tab:axes; ~3 min)
PYTHONPATH=src python scripts/continuous_smooth_probe.py
                        # smooth-learner table (tab:smooth; ~11 s)
```

```

PYTHONPATH=src python scripts/continuous_pendulum.py
# pendulum table (tab:pendulum; ~2 min)
PYTHONPATH=src python scripts/continuous_cem.py
# second planner family (tab:cem; ~29 min)
PYTHONPATH=src python scripts/continuous_patch2d.py
# PatchField2D mechanism (tab:patch2d)
PYTHONPATH=src python scripts/continuous_cem_patch2d.py
# PatchField2D CEM rows (sec:cem; ~14 min)
PYTHONPATH=src python scripts/continuous_eps_sweep_patch2d.py
# PatchField2D eps rows (tab:eps-sweep text; ~8 min)
PYTHONPATH=src python scripts/continuous_mitigation_patch2d.py
# PatchField2D mitigation (tab:patch2d-mitigation)
python scripts/make_paper2_figures.py # figures from the JSONs

```

LLM arms (Azure credentials in .env; the runbook — commands, cost, predictions, troubleshooting — is the section “Runbook — LLM arms” of the design spec under docs/specs/, dated 2026-07-06):

```

PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 5
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 5 --x-wall 4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 5
# Pendulum synthesis arm (second instrument; Azure credentials as above)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
--instrument pendulum --th-stop 1.4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
--instrument pendulum --th-stop 1.4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
--instrument pendulum --th-stop 1.0
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
--instrument pendulum --th-stop 1.0
# Qwen cross-family spot-checks
# (needs HF_TOKEN: HF Inference Providers router)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 3 \
--compat-model "Qwen/Qwen3-Coder-30B-A3B-Instruct"
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 3 \
--instrument pendulum --th-stop 1.4 \
--compat-model "Qwen/Qwen3-Coder-30B-A3B-Instruct"
# Claude agent-relayed cross-family spot-checks (agent scaffold relays
# pipeline-identical messages; see scripts/continuous_claude_step.py,
# results/continuous_claude_relay.json + results/claude_relay_transcripts/).
# One relay round = init (writes msgN.txt) -> relay it -> check (gates the
# reply, writes msgN+1.txt or classifies). The 2D arm, per seed:
PYTHONPATH=src python scripts/continuous_claude_step.py init 10000 \
results/claude_relay_transcripts --instrument patch2d --k1 3 --k2 7 \
--arm incomplete
PYTHONPATH=src python scripts/continuous_claude_step.py check 10000 \
results/claude_relay_transcripts 0 REPLYFILE --instrument patch2d \
--k1 3 --k2 7 --arm incomplete --model-label "claude-sonnet-5 (agent-relayed)"

```



```

# Per-iteration ledger, re-derived from the versioned transcripts (CPU only):
PYTHONPATH=src python scripts/claude_relay_ledger.py \
    results/claude_relay_transcripts --tag patch2d_k3_7 --k1 3 --k2 7
# PatchField2D synthesis arm (4D bi-modal; two bi-knob cells)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
    --instrument patch2d --k1 3 --k2 7
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
    --instrument patch2d --k1 3 --k2 7
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
    --instrument patch2d --k1 5 --k2 9
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
    --instrument patch2d --k1 5 --k2 9
# Mitigation sweep (CPU only)
PYTHONPATH=src python scripts/continuous_mitigation.py
# eps-sensitivity sweep (CPU only)
PYTHONPATH=src python scripts/continuous_eps_sweep.py

```

## References

- Javier Aguilar Martín. When a verified world model still loses: Play-adequacy vs prediction-accuracy in LLM-synthesized code world models, 2026. Companion paper (paper 1). arXiv:2607.14169, <https://arxiv.org/abs/2607.14169>; repository: <https://github.com/JaviMaligno/code-world-models>.
- Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI)*, pages 2669–2678, 2018.
- Yashwanth Annpureddy, Che Liu, Georgios Fainekos, and Sriram Sankaranarayanan. S-TaLiRo: A tool for temporal logic falsification for hybrid systems. In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, volume 6605 of *Lecture Notes in Computer Science*, pages 254–257. Springer, 2011.
- Alberto Bemporad and Manfred Morari. Control of systems integrating logic, dynamics, and constraints. *Automatica*, 35(3):407–427, 1999.
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Koen Claessen and John Hughes. QuickCheck: A lightweight tool for random testing of Haskell programs. In *Proceedings of the Fifth ACM SIGPLAN International Conference on Functional Programming (ICFP)*, pages 268–279, 2000.
- Anthony Corso, Robert J. Moss, Mark Koren, Ritchie Lee, and Mykel J. Kochenderfer. A survey of algorithms for black-box safety validation of cyber-physical systems. *Journal of Artificial Intelligence Research*, 72:377–428, 2021.
- Rémi Coulom. Efficient selectivity and backup operators in monte-carlo tree search. In *Computers and Games (CG 2006)*, volume 4630 of *Lecture Notes in Computer Science*, pages 72–83. Springer, 2006.

- Nima Fazeli, Roman Kolbert, Russ Tedrake, and Alberto Rodriguez. Parameter and contact force estimation of planar rigid-bodies undergoing frictional contact. *The International Journal of Robotics Research*, 36(13–14):1437–1454, 2017.
- Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 10764–10799, 2023.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Levente Kocsis and Csaba Szepesvári. Bandit based Monte-Carlo planning. In *Proceedings of the 17th European Conference on Machine Learning (ECML)*, pages 282–293, 2006.
- Nathan O. Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control (L4DC)*, volume 120 of *Proceedings of Machine Learning Research*, pages 761–770, 2020. arXiv:2002.04523.
- Wolfgang Lehrach, Daniel Hennes, Miguel Lazaro-Gredilla, Xinghua Lou, Carter Wendelken, Zun Li, Antoine Dedieu, Jordi Grau-Moya, Marc Lanctot, Atil Iscen, John Schultz, Marcus Chiam, Ian Gemp, Piotr Zielinski, Satinder Singh, and Kevin P. Murphy. Code World Models for general game playing. In *International Conference on Learning Representations (ICLR)*, 2026. Poster; OpenReview id 1UoB7IWiku.
- Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 9493–9500, 2023.
- Anusha Nagabandi, Gregory Kahn, Ronald S. Fearing, and Sergey Levine. Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.
- Simone Paoletti, Aleksandar Lj. Juloski, Giancarlo Ferrari-Trecate, and René Vidal. Identification of hybrid systems: A tutorial. *European Journal of Control*, 13(2-3):242–260, 2007.
- James B. Rawlings, David Q. Mayne, and Moritz M. Diehl. *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing, 2nd edition, 2017.
- Reuven Y. Rubinstein and Dirk P. Kroese. *Simulation and the Monte Carlo Method*. Wiley, 3rd edition, 2017.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.